

Course Orientation & Data Science Process

What data science is and how a project runs

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Learning objectives

By the end of this class, students should be able to

- explain what data science is, using a defensible definition, and say which disciplines it draws on;
- describe the kinds of data a data scientist meets, and state precisely what the rows and columns of a tabular data set represent;
- distinguish exploratory, predictive, and descriptive analytics, and recognize each in a worked example on the same data set;
- name and order the six stages of CRISP-DM, and state the question each stage answers;
- explain why the process is a loop rather than a pipeline; and
- locate the course's assessment, deadlines, communication channels, and required tools.

The class moves in four steps. It first motivates the field from the labour market and from the growth in data and computing. It then introduces data science through one clinical data set, using that data set to illustrate the three task families. It then presents CRISP-DM as the standard project life cycle. It closes with the course's own structure, which follows the same life cycle.

1 Why learn data science

1.1 The labour market

A ranking of the best jobs in Canada illustrates the landscape in Table 1. This is a historical example of how posting growth and salary describe different aspects of a job market, not a current employment forecast.

Table 1. Historical 2022 ranking of data-related jobs in Canada by posting growth

Rank	Job	Average salary	Growth (%) in postings, 2019–22
1	Data engineer	\$85,706	242
2	Site reliability engineer	\$90,010	218
3	Engineering manager	\$111,515	161
...			
19	Data scientist	\$76,938	60

Source: Canadian HR Reporter. Data engineering, first on this list, is the work of making data available and reliable. Much of what this course calls data preparation is the analyst's share of that work.

1.2 What is driving the field

Increasing volumes of data. Cheap and varied sensors, wider networking, and the Internet of Things all generate records that are stored by default rather than discarded.

Increasing computing power. Methods that were once impractical on a single machine now run on a laptop, which changes what is worth attempting.

Data science is also an *interdisciplinary* field with application across many domains: healthcare; finance and banking; transportation and smart cities; business and e-commerce; social media and social network analysis; and sustainability and climate. Each of these supplies both the questions and the domain knowledge without which an analysis cannot be interpreted.

Discussion prompt from the class

Why did you enrol in this course, and what do you hope to learn? Later in the class: now that you know a bit more about what data science can do, what application areas seem most interesting to you?

2 Data and tabular data

2.1 Types of data

Volumes are increasing across several different types of data.

- Tabular data.
- Databases, that is, multiple related data tables.
- Text.
- Multimedia: images, video, sound, and so on.
- Data streams and time series.

This course works with **tabular data**. That restriction is deliberate. Almost every idea in the course, from distance to model assessment, is defined first on a table, and the definitions transfer once the other data types have been given a tabular representation.

2.2 The structure of a table

Key terminology follows.

- A data set is a collection of observations.
- A data matrix of dimensions $n \times p$ contains n observations or cases, each described by p variables.
- Rows represent the cases. They are also called objects, examples, instances, or records.
- Columns represent variables. They are also called features, fields, or attributes.

Identify the observational unit

The observational unit is what one row describes. It might be one patient, one household, or one sensor reading. State it before counting observations. A numeric variable measures a quantity; a categorical variable records a label. An identifier names a record and should not be averaged.

2.3 The example data set

The lecture's running example is the Heart Disease data set: **12 attributes** and **918 cases**, from the UCI Machine Learning Repository, distributed through Kaggle. The first six rows are shown in Table 2.

Download and extract `heart.csv` from the Kaggle dataset page. Place the CSV in your working directory, the folder from which your R or Python session reads files. The import demonstration below

inspects the table. The later model figures preview questions we will learn to solve during the course; you do not need to fit those models this week.

Table 2. First six rows of the Heart Disease data set

Row	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS
	numeric	category	category	numeric	numeric	category
1	40	M	ATA	140	289	0
2	49	F	NAP	160	180	0
3	37	M	ATA	130	283	0
4	48	F	ASY	138	214	0
5	54	M	NAP	150	195	0
6	39	M	NAP	120	339	0

Row	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
	category	numeric	category	numeric	category	category
1	Normal	172	N	0.0	Up	0
2	Normal	156	N	1.0	Flat	1
3	ST	98	N	0.0	Up	0
4	Normal	108	Y	1.5	Flat	1
5	Normal	122	N	0.0	Up	0
6	Normal	170	N	0.0	Up	0

One row is one patient. Seven variables are categorical and five are numeric, including the four integer-valued measurements and Oldpeak. These are analytical roles, not the storage types returned by an import function. FastingBS and HeartDisease encode categories as integers. HeartDisease is the outcome that the predictive example will try to estimate; in the full table, 508 of the 918 patients have it, which is the 55% that appears at the root of the decision tree later in this class.

R

```
heart <- read.csv(
  "heart.csv",
  stringsAsFactors = TRUE
)

dim(heart)      # 918 12
head(heart)
str(heart)
table(heart$HeartDisease)
```

Python

```
import pandas as pd

heart = pd.read_csv("heart.csv")

heart.shape      # (918, 12)
heart.head()
heart.dtypes
heart["HeartDisease"].value_counts()
```

Both imports produce 918 rows and 12 columns. In R, <- assigns a value to a name; in Python, = does that job. R's dim and Python's shape report the dimensions. The head calls show the first rows; str and dtypes inspect storage types. The final calls count the two outcome categories. R's \$ and Python's bracket notation select a column. R imports text as factors here, its representation of categories. Integer-coded labels still need to be identified from the dataset documentation.

3 What data science is

The lecture adopts a definition from the data mining literature.

Definition

“Data mining is the analysis of (often large) observational data sets to find unsuspected relationships and to summarize the data in novel ways that are both understandable and useful to the data owner.”

Hand, Mannila, and Smyth, *Principles of Data Mining*, MIT Press, 2001, page 1.

This definition emphasizes discovering relationships and producing useful summaries. Data science combines that work with computing, statistical reasoning, and knowledge of the subject being studied. **Observational** means the researcher does not assign an experimental intervention. Such data may be collected for a research question; an observed relationship alone does not establish cause and effect. Data science also uses experiments.

The lecture pairs the definition with a three-circle diagram. Data science sits at the intersection of computer science and information technology, mathematics and statistics, and domain or business knowledge. The pairwise intersections are themselves recognizable fields. Machine learning sits between computing and statistics, software development between computing and the domain, and traditional research between statistics and the domain.

4 Data science tasks

The lecture divides the work into three task families, summarized in Table 3.

Table 3. Three data science task families and the questions they answer

Family	Sub-tasks	Question it answers
Exploratory data analysis	summarization, visualization	What is in this table?
Predictive analytics	classification, regression	What value should I expect for an unknown case?
Descriptive analytics	clustering, anomaly detection	What structure is present, without a target?

The next figures illustrate all three questions on the Heart Disease data. Reading the output is the objective today. Model construction belongs to later weeks; Lab 01 uses descriptive statistics and plots only.

4.1 Exploratory data analysis: an example

Figure 1 illustrates the process. The left panel is a histogram of Age, stacked by HeartDisease, with the median age marked. The second is a scatter plot of MaxHR against Age, coloured by HeartDisease. Together they say that patients with heart disease in this sample tend to be older and to reach a lower maximum heart rate. Nothing has been fitted; the figures describe the sample.

In a histogram, each bar counts observations in an interval, called a bin. A scatter plot places one point at each pair of measurements. The median is the middle value after sorting, or the mean of the two middle values for an even number of observations. Class B develops these summaries and shows how to make and save simple plots.

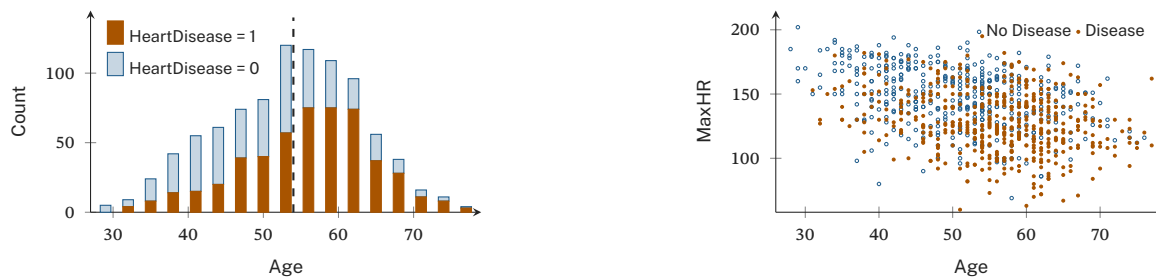


Figure 1. Exploratory views of all 918 patients. Three-year age bins start at 27.5; the dashed line marks the median age of 54. The scatter uses unjittered observations, so coincident points overlap. Neither plot fits a model.

4.2 Predictive analytics: an example

A model is a rule fitted to data. Classification predicts a category, called the target. A classification tree expresses its rules as branches. A tree with HeartDisease as the target appears in Figure 2. Each node in the printed tree carries three values: the predicted class, the proportion of disease cases in the node, and the share of the training data that reaches the node.

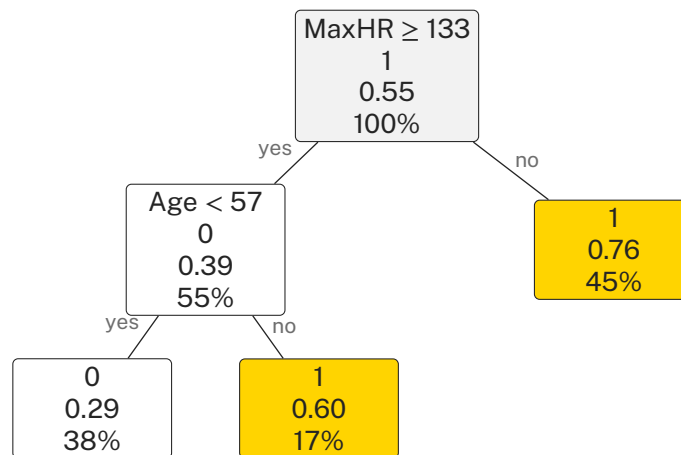


Figure 2. A depth-two classification tree for HeartDisease. Each node gives the predicted class, the proportion of disease cases, and the share of the training data reaching it. Shaded leaves predict disease.

Read the tree as a set of rules. A patient whose maximum heart rate is below 133 is predicted to have heart disease, and 76% of the patients in that leaf do. Above that threshold, age decides. Below 57 the prediction is no disease, at 57 or above it is disease. The two splits cut the Age–MaxHR plane into three rectangles, shown in Figure 3. The fitted boundaries are 132.5 and 56.5, equivalent to these integer rules on the recorded measurements. The training accuracy is **71.24%**, or 654 correct predictions out of 918.

The 918 patients used to fit the rules are the training data. The reported accuracy describes agreement on those same patients; it does not estimate performance on new patients. Tree fitting is taught in Week 8 and performance estimation in Week 9.

4.3 Descriptive analytics: an example

Clustering groups observations with similar measurements, without using a target category. In this preview, software groups patients using age and maximum heart rate, then marks five that lie far from their group’s centre. The gold markers identify CSV rows 830, 3, 508, 391, and 569, excluding the header. Figure 4 shows where those observations lie.

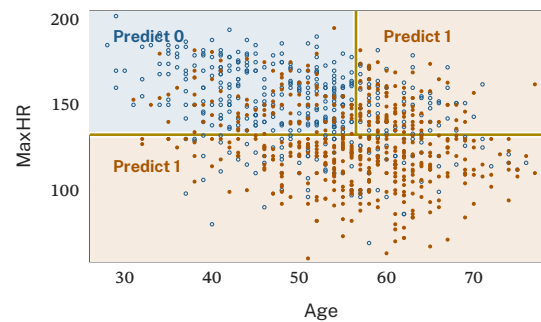


Figure 3. The same fitted tree partitions the Age–MaxHR plane. Blue regions predict no disease and rust regions predict disease; the points show the observed classes. The gold boundaries represent the two fitted splits.

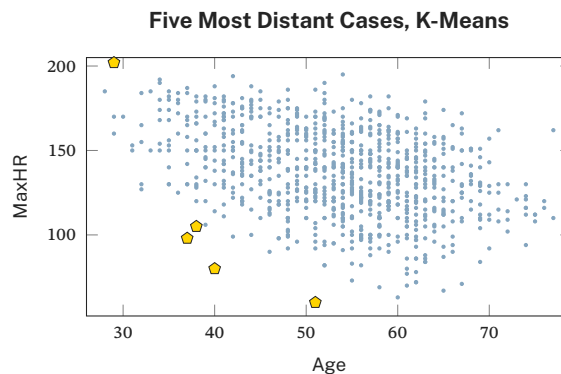


Figure 4. Five observations selected for investigation are marked with gold pentagons. The axes retain the original measurement units. Clustering and distance-based screening are taught in Weeks 11–12.

Interpretation guidance

A flagged case is not a wrong case. Anomaly detection produces a ranked list for human attention. Whether an unusual maximum heart rate is a data-entry error, a rare patient, or the most clinically interesting record in the table is a question the method cannot answer.

Check your understanding

Which example uses HeartDisease as the category to predict? If that column were removed, could you still plot age against maximum heart rate and inspect unusual patients?

5 CRISP-DM: the typical data science process

CRISP-DM is the **C**Ross **I**ndustry **S**tandard **P**rocess for **D**ata **M**ining. In the Data Science Process Alliance’s 2020 convenience poll of 109 respondents, CRISP-DM received 49%, followed by Scrum, Kanban, and “my own” at 18%, 12%, and 12%. The remaining responses were Microsoft TDSP (4%), Other (3%), None (2%), and SEMMA (1%). The eight reported percentages sum to 101% after rounding. This self-selected poll does not estimate prevalence among all practitioners.

The six stages and their associated questions, shown in Figure 5, are as follows.

1. **Business understanding.** What are the business goals? How can data science help achieve them?
2. **Data understanding.** What data do we have or need? Are there data quality issues?
3. **Data preparation.** How do we clean and transform data for modelling?

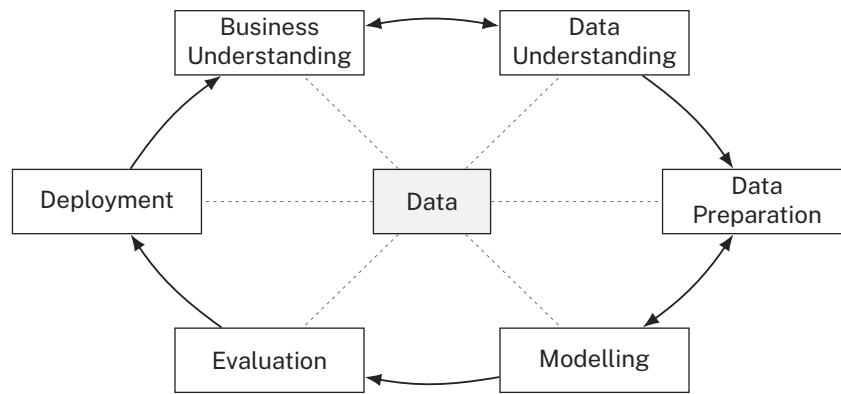


Figure 5. The six CRISP-DM stages. The cycle runs clockwise from business understanding, the two double-headed arrows mark the pairs that are worked back and forth, and every stage reads from and writes to the same data.

4. **Modelling.** What modelling techniques should we use? How do we calibrate them?
5. **Evaluation.** Does the selected solution meet the business goals?
6. **Deployment.** How do we deliver our solution to the business?

Two features of the diagram carry most of the meaning. The double-headed arrows between business and data understanding, and between data preparation and modelling, record that these pairs are negotiated together, because the choice of model determines what the prepared table must look like, and what the data can support constrains what goal is realistic. The outer ring records that the process repeats; evaluation returns to business understanding rather than terminating the project.

Discussion prompts from the class

At the data understanding stage: can you think of any potential data quality issues or limitations? At the deployment stage: what kind of deliverables would provide value to stakeholders?

Think, pair, share

Take a minute to think about a concrete data science application you are curious about. What data would you need, and do you think it would be readily available? What challenges do you expect to face? Discuss with your neighbour, then volunteer to share with the class.

6 About the course

6.1 The syllabus map

The tentative syllabus is drawn as one map whose spine is CRISP-DM. Business understanding, data understanding, data preparation, data modelling, evaluation, and deployment run left to right, with evaluation feeding back to business understanding. Three of those stages carry the course's technical content, listed in Table 4.

6.2 Learning outcomes for the course

By the end of the course, you should be able to

1. describe the main steps in a data science project;
2. transform data to prepare it for analysis and modelling;

Table 4. Course topics mapped to CRISP-DM stages

Stage	Topics hanging from it
Data understanding	importing data; summarizing; visualizing
Data preparation	cleaning; transforming; feature engineering; reducing dimensionality
Data modelling, predictive	tasks: classification, regression. Models: linear models, support vector machines, tree-based models. Assessment: performance metrics, performance estimation methods
Data modelling, descriptive	distance functions; clustering, split into partition-based and hierarchical clustering; outlier detection
Deployment	reproducible reports; optional Shiny applications in R or Python

3. explore data using appropriate statistics and visualizations;
4. design and implement a data science application, using predictive and descriptive modelling techniques;
5. evaluate, compare, and select models for a dataset; and
6. prepare reproducible reports in R Markdown or Python notebooks.

6.3 Technologies

Prepare one working R or Python environment. A free cloud notebook is sufficient, using Posit Cloud or Kaggle for R, or Posit Cloud, Kaggle, or Google Colab for Python. For local work, install R from CRAN with an editor such as RStudio, or Python 3 with a virtual environment and an editor such as VS Code or JupyterLab. Both languages use the same assessment requirements and rubric. Contact the instructor early if computer access is a barrier. The syllabus's reporting outcome is R Markdown or Python notebooks. Shiny applications in either language are an optional extension, not an additional assessed requirement.

Week 1 uses base R, or Python with pandas and matplotlib. Later classes introduce their additional packages before using them. Free cloud services have usage caps. Save your files regularly and record the dependencies needed to reproduce the analysis. Keep assessed notebooks private until the deadline and the instructor opens sharing.

6.4 Communication

- Announcements are posted in the course GitHub repository, <https://github.com/MAPS-Lab/CSCI-3141>. Course materials are public and can be read without signing in. You need a GitHub account for your assessment portfolio. Check the repository and your institutional email regularly; the assessment section below explains submission and sharing.
- Use Microsoft Teams for class questions and office-hour appointments.
- For private matters, use e-mail, with a subject line of the form [CSCI 3141] Subject matter.
- Instructor is Prof. Gabriel Spadon, spadon@dal.ca. Replies are normally within three business days.

Office hours are held **by appointment through Microsoft Teams**. Write ahead with the subject you would like to discuss. Individual and small-group appointments are available. Bring a focused question and ask early rather than waiting until a deadline.

6.5 Classes

Classes run Tuesdays and Thursdays, **16:05–17:25**, in Carleton Tupper Building room **3H01**, in person, across the twelve teaching weeks of the term. The syllabus and the course calendar carry this term's dates. Classes include in-class response questions and work in pairs or small groups, which do not reproduce well from slides alone. Course materials are published in the course GitHub repository. There is no separate lab section; the integrated lab work begins in the Thursday classes.

6.6 Academic integrity and conduct

Academic integrity is expected from all students at Dalhousie. Review the Academic Integrity resources linked from the course GitHub repository. Originality checking follows syllabus A.9.1. Everyone at Dalhousie should be treated with dignity and respect, in class and online.

Generative AI

Generative AI is **prohibited on integrated labs** and examinations, and on assignments unless its use is expressly authorized in the assignment instructions. When it is authorized, follow the stated disclosure requirements.

6.7 Assessment

Table 5 lists the course assessment components, their weights, and the rule applied to each.

Table 5. Course assessment components and their weights

Component	Weight	Rule
Integrated labs	15%	best 10 of 12 results
Assignment 1	7.5%	import data and perform basic data exploration; due 11 October 2026
Assignment 2	7.5%	pre-process and explore data; due 1 November 2026
Assignment 3	15%	design and implement a predictive modelling solution; due 29 November 2026
Assignment 4	15%	design and implement a descriptive modelling solution; due 9 December 2026
Midterm	15%	in class, 20 October 2026
Final examination	25%	cumulative, scheduled by the Registrar
Total	100%	

The weighted total, the combined assignment component, and the final examination must each reach 50%. The syllabus gives the full passing requirements.

Integrated labs are marked with the common course rubric, and the best 10 of 12 results contribute 15% of the course grade. Submit each lab by 23:59 two calendar days after it begins. Assignments are individual work, due at 23:59 on the announced date. Assignments 1 to 3 lose **5 percentage points per late calendar day**, up to seven days, after which later work requires an approved absence or accommodation process. Assignment 4 has no routine late window. Use the syllabus's equivalent private repository when accommodation, safety, privacy, copyright, or licensing requires it.

The midterm takes place in class on October 20, from 16:05 to 17:25, and the Registrar schedules the cumulative final examination. Bring valid Dalhousie photo ID. Course and personal notes are allowed unless the examination lists narrower approved aids. Internet access, communication, and generative AI are prohibited.

Build the portfolio in a private work repository with instructor access. Commit regularly, then submit the full commit SHA and its commit URL through the announced private channel by the deadline. A branch link or movable tag alone does not fix the submitted version. Keep cloud notebooks and answer-revealing links private too. After the deadline and when the instructor opens sharing, publish that same commit and its history in your public portfolio and share that version in class.

Preserve assessment history permanently. Do not amend, rebase, squash, delete, replace, force-push, or backdate recorded commits, and do not move submitted tags. Keep submitted commits reachable. Record corrections as new commits, retaining the original submission. Git author dates alone are not deadline evidence; the submission receipt and instructor-retained snapshot provide the independent reference. Contact the instructor if an accidental disclosure requires correction.

Course materials released by the instructor may be shared with credit to the author and CSCI 3141 unless explicitly marked private or restricted. Preserve source links, third-party attributions, and licence terms. This permission does not authorize sharing another student's work or releasing assessed work early.

For an eligible short-term absence, contact the instructor before the requirement and submit the declaration within three calendar days after the last day of the absence, at most twice per course. Follow syllabus A.7.3 for other absences.

6.8 Dates

Table 6 lists the key course dates and deadlines for fall 2026.

Table 6. Key course dates and deadlines for fall 2026

Date, 2026	Event	Kind
8 September	Classes begin	course
22 September	Last day to add	University
7 October	Last day to drop without a W	University
11 October	Assignment 1 due	course
20 October	Midterm, in class	course
1 November	Assignment 2 due	course
5 November	Last day to drop with a W	University
9–13 November	Fall Break	University
29 November	Assignment 3 due	course
3 December	Last class	course
9 December	Assignment 4 due	course
Registrar	Final examination	University

6.9 Course text and resources

There is **no required paid textbook**. The lecture slides, notes, and integrated lab handouts in the course GitHub repository are the primary material. Cheatsheets and documentation links are posted with each module. The following texts are optional further reading. Check Dalhousie Libraries for access before obtaining a copy.

Rigdon, S. E., Fricker, R. D., and Montgomery, D. C., *Introduction to Probability and Statistics for Data Science with R*, Cambridge University Press, 2024.

Shah, C., *A Hands-On Introduction to Data Science with Python*, 2nd edition, Cambridge University Press, 2026.

Torgo, L., *Data Mining with R: Learning with Case Studies*, 2nd edition, CRC Press, 2017. Earlier editions are acceptable.

6.10 This week

Read the syllabus, check course repository and Teams access, and prepare one working local or cloud environment in R or Python before Thursday. Thursday starts Lab 01. The class begins with a summary-statistics warm-up, then you choose one listed Atlantic environmental dataset and establish its descriptive baseline.

7 Summary

Data science combines computing, statistics, and domain knowledge to learn from observational and experimental data. This course uses tabular data, with n cases as rows and p attributes as columns. The Heart Disease example has 918 cases and 12 attributes.

Exploratory analysis summarizes and visualizes the sample. Predictive analytics learns a target, as in the tree splitting MaxHR at 133 and Age at 57. Its 71.24% training accuracy does not establish future performance. Descriptive analytics finds structure without a target, as in the k-means example that marks five patients for investigation. Those models are course previews; this week you will import a table and explain its measurements, summaries, and plots.

CRISP-DM connects business understanding, data understanding, data preparation, modelling, evaluation, and deployment. Later evidence can revise earlier decisions. The course follows this process, and its four assignments move from import to prepare to predict to describe. Consult the syllabus for assessment and integrity rules, and keep portfolio work private until its deadline and the instructor's release.